

ANSI/ASNT CP-189-2024
American National Standard

ASNT STANDARD

FOR QUALIFICATION AND
CERTIFICATION OF NONDESTRUCTIVE
TESTING PERSONNEL

2024

EDITION



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ASNT Mission Statement:
ASNT's mission is to advance the field of nondestructive testing.

ASNT Code of Ethics:
The *ASNT Code of Ethics* was developed to provide members of the Society with broad ethical statements to guide their professional lives. In spirit and in word, each ASNT member is responsible for knowing and adhering to the values and standards set forth in the Society's Code. More information, as well as the complete version of the *Code of Ethics*, can be found on ASNT's website, *asnt.org*.

ANSI/ASNT CP-189-2024
American National Standard
ASNT Standard for Qualification and
Certification of Nondestructive Testing Personnel

The American Society for Nondestructive Testing Inc.
Approved 23 June 2023
American National Standards Institute

Abstract

This standard applies to personnel whose specific tasks or jobs require appropriate knowledge of the technical principles underlying nondestructive testing (NDT) methods for which they have responsibilities within the scope of their employment. These specific tasks or jobs include, but are not limited to, performing, specifying, reviewing, monitoring, supervising, and evaluating NDT work.

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Employers or other persons utilizing NDT services are cautioned that they retain full responsibility for the ultimate determination of the qualifications of NDT personnel and for the certification process. The process of personnel qualification and certification as detailed in the standard does not relieve the employer of the ultimate legal responsibility to ensure that the NDT personnel are fully qualified for the tasks being undertaken.

This standard is subject to revision or withdrawal at any time by ASNT.

AMERICAN NATIONAL STANDARD

American National Standard Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria for approval have been met by the standards developer.

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FOREWORD

(Not a part of American National Standard CP-189-2024)

An essential element in the effectiveness of nondestructive testing (NDT) is the qualification of the personnel who are responsible for and who perform NDT. Formal training and actual experience are important and necessary elements in acquiring the skills necessary to effectively perform nondestructive tests.

The American Society for Nondestructive Testing Inc. (ASNT) has, therefore, undertaken the preparation and publication of this standard, which specifies the procedures, essential factors, and minimum requirements for qualifying and certifying NDT personnel.

The *ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel* (Document No. ASNT CP-189) was initially processed and approved for submittal to the American National Standards Institute (ANSI) by the ASNT Standards Writing Committee. This revision was processed by the ASNT Standards Council. Council approval of the standard does not necessarily imply that all committee members voted for its approval. At the time it approved this standard, the Standards Council had the following members:

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Approved by the Standards Council 23 June 2023.

Appendixes C and D are not part of this American National Standard.

Notice to users of this document (Not a part of American National Standard CP-189-2024):

- ▶ When certificates or wallet cards are issued and provided by the employer, the requirements of ASNT Policy G-14 shall be followed to ensure appropriate usage of the ASNT name, acronym, and logo.
- ▶ The certificate shall indicate that the certification is in accordance with the requirements of the employer's certification procedure.
- ▶ See paragraph 9.2.7 for further guidance on properly issued certification certificates and wallet cards.

Ethics Statement

The NDT profession plays a critical role in ensuring public safety by providing information on the condition and integrity of critical infrastructure and products through its various NDT methods and techniques. The NDT profession has a responsibility to act ethically in all activities including the proper collection, interpretation, and reporting of NDT data. To this end, ASNT highly encourages all individuals and companies using this standard/document to incorporate the topic of ethics and ethical behavior of its employees and contractors into their training programs and examinations.

SUMMARY OF CHANGES

The changes made are also indicated with a line in the margin for easy identification.

CP-189 2024 Edition

1. Paragraphs 1.5, 2.2.1, 6.3.2.1, and 6.3.3.3. These paragraphs have been modified to require the employer to assess the ability of the referenced exams to meet the employer's requirements and to document that acceptance.
2. Paragraphs 2.1.15 and 2.1.21. The revision to 2.1.21 was to further clarify the term *predictive maintenance*. Additionally, it was determined necessary to show contrast to the term NDT, so the definition for *nondestructive testing* from ASTM E1316 was included as well (new paragraph 2.1.15).
3. Note after paragraph 2.2.1. To clarify how Level III personnel currently certified by ASNT NDT Level III in neutron radiography or vibration analysis can still utilize the standard if they maintain their certification, even though ASNT has stopped offering these exams to new applicants.
4. Paragraphs 2.2.1, 6.3.1.1, 6.3.2.1, and 6.3.3.3. Added new ASNT 9712 Program. ACCP has been left in for now, since there is five-year transition in place.
5. Paragraph 6.1.1. Restriction added for use of eye drops to temporarily improve vision. This issue and some other vision-related topics are in the hands of a Technical Advisory Group (TAG) Committee, so more changes are coming at a later date, including more on the eye drops.
6. Paragraph 6.3.1. Added use of ACCP, ASNT NDT, or ASNT 9172 General examination being used to fulfill General examination criteria.
7. Paragraph 6.4.1.3. Requirement added to ensure personnel administering written multiple-choice examinations are trained and the training is documented.
8. Paragraph 6.8. Added new section to address the "Certification" process.
9. Table 1. Added digital radiography (DR), computed radiography (CR), and computed tomography (CT) image interpreter requirements.
10. Paragraph 9.2.2. Added dates of when certification actually begins. This issue came out of an interpretation inquiry to SNT-TC-1A about the date certification starts. See Interpretation 85-6.
11. Paragraph 9.2.1.1. Added "name of certified individual."
12. Appendix B. Added DR, CR, and CT image interpreter requirements.

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ASNT STANDARD FOR QUALIFICATION AND CERTIFICATION OF NONDESTRUCTIVE TESTING

1.0 Scope

- 1.1 This standard establishes the minimum requirements for the qualification and certification of nondestructive testing (NDT) and predictive maintenance (PdM) personnel.
- 1.2 This standard details the minimum training, education, and experience requirements for NDT personnel and provides criteria for documenting qualifications and certification.
- 1.3 This standard requires the employer to establish a procedure for the certification of NDT personnel.
- 1.4 This standard requires that the employer incorporate any unique or additional requirements in the certification procedure.

2.0 Definitions

- 2.1 Purpose. These definitions are intended to clarify the meanings of terms used in this standard, as they apply to this standard, and only to this standard. No broader application of these definitions is implied. For specific definitions related to NDT, see paragraph 10.2.
- 2.1.1 **Calibration, Instrument:** the comparison of an instrument with, or the adjustment of an instrument to, a known reference(s) often traceable to the applicable country's national institute or standards body. (See also Standardization, Instrument.)
- 2.1.2 **Certification:** written testimony that an individual has met the applicable requirements of this standard.
- 2.1.3 **Certification Procedure:** a written procedure, developed by the employer, that details the requirements for qualification and certification of an employee to this national standard.
- 2.1.4 **Certifying Authority:** the person or persons properly designated in the certification procedure to sign certifications on behalf of the employer.
- 2.1.5 **Closed-book Examination:** an examination administered without access to reference material except that supplied with or in the examination.
- 2.1.6 **Documented:** the condition of being in written form.
- 2.1.7 **Education:** an institutionalized program, prescribed by appropriate authorities, that is offered by schools, institutes, organizations, colleges, or universities established for the sole purpose of providing instruction in an orderly, planned, and systematic fashion.
- 2.1.8 **Employer:** the corporate, private, or public entity that employs personnel directly or indirectly for wages, salary, fees, or other considerations. This would include organizations that obtain their qualified supplemental workforce personnel through third-party agencies, providing the use and certification of those supplemental employees is addressed in the employer's certification procedure.
- 2.1.9 **Examination:** a formal, controlled, documented assessment of knowledge or skills conducted in accordance with a procedure.
- 2.1.10 **Experience:** actual performance of an NDT method conducted in the work environment resulting in the acquisition of knowledge and skill. This does not include formal classroom training but may include laboratory and on-the-job training as defined by the employer's certification procedure.
- 2.1.11 **Grading Unit:** a qualification specimen can be divided into sections called grading units, which do not have to be equal length or be equally spaced. Grading units are unflawed or flawed and the percentage of flawed/unflawed grading units required shall be approved by the NDT Level III.
- 2.1.12 **Formal Training:** an organized and documented program of activities designed to impart the knowledge and skills required to be qualified to this standard. Formal training may be a mix of classroom, practical, and programmed self-instruction as approved by the responsible NDT Level III.

- 2.1.13 **General Examination:** a written examination addressing the basic principles of the applicable NDT method.
- 2.1.14 **Method:** one of the disciplines of NDT: for example, ultrasonic testing, within which various test techniques may exist.
- 2.1.15 **Nondestructive Testing (NDT):** the development and application of technical methods to examine materials or components in ways that do not impair future usefulness and serviceability in order to detect, locate, measure, and evaluate flaws; to assess integrity, properties, and composition; and to measure geometrical characteristics.
- 2.1.16 **NDT Instructor:** an individual qualified and designated in accordance with this standard to train or educate NDT personnel. (See also paragraph 3.7.)
- 2.1.17 **NDT Procedure:** a written instruction for conducting a nondestructive test.
- 2.1.18 **Outside Organization:** an agency or individual who provides NDT Level III services. (See also paragraph 4.5.)
- 2.1.19 **Personalized Instruction:** personalized instruction may consist of blended classroom, supervised laboratory, and/or hybrid online competency-based course delivery. Modular content is covered through online presentations, in the classroom, and/or in small groups. Personalized instruction also enables students to achieve competency using strategies that align with their knowledge, skills, and learning styles.
- 2.1.20 **Practical Examination:** an examination used to demonstrate an individual's ability in conducting the NDT methods that will be performed for the employer. For practical examinations, questions and answers need not necessarily be written, but observations and results must be documented.
- 2.1.21 **Predictive Maintenance (PdM):** evaluates the condition of equipment (typically in service) by performing periodic or continuous (online) equipment condition monitoring. Condition monitoring evaluates leading performance indicators of specified machinery, components, and assemblies (including items such as buildings; e.g., structures in whole and in part), detectable via the applied method. PdM uses principles of statistical process control to assess the condition by focusing on leading indicators that may signify deterioration in performance or potential equipment, structure, or material failure. The ultimate goal of PdM is to perform maintenance based upon the condition such that the maintenance activity is cost-effective and before the equipment loses optimum performance or fails.
- 2.1.22 **Qualification:** the education, skills, training, knowledge, and experience required for personnel to properly perform to a specified NDT level.
- 2.1.23 **Specific Examination:** a written examination to determine an individual's understanding of procedures, codes, standards, specifications, and equipment or instrumentation for an NDT method used by the employer.
- 2.1.24 **Standardization, Instrument:** the adjustment of an NDT instrument using an appropriate reference standard to obtain or establish a known and reproducible response. (This is usually done prior to performing any NDT, but can be carried out anytime there is concern about the performance of the examination or instrument response.) (See also Calibration, Instrument.)
- 2.1.25 **Test Technique:** a category within an NDT method: for example, immersion ultrasonic testing.
- 2.2 **NDT Levels**
 - 2.2.1 **NDT Level III.** An individual possessing a currently valid ASNT NDT, PdM, ACCP Professional (see paragraphs 10.1.2, 10.1.3, or 10.1.5), or ASNT 9712 Level III certificate and certified in accordance with this standard. (See also Section 3.0.) Reference to ASNT Level III throughout this standard implies the individual holds one of these certificates.
NOTE: As of the printing of this standard, the ASNT NDT Level III neutron radiography (NR) and vibration analysis (VA) examinations are no longer being offered by ASNT. Personnel who hold these certificates are still certified as long as they maintain the certification with ASNT through the recertification by points process.
 - 2.2.2 **NDT Level I, NDT Level II.** An individual certified in accordance with this standard. (See also Section 3.0.)

3.0 Levels of Qualification

- 3.1 **Classification.** Six levels of qualification are defined in terms of the skills and knowledge required in a given method or methods to perform specified NDT activities.
- 3.2 **NDT Level III.** An NDT Level III shall have the skills and knowledge to establish techniques; to interpret codes, standards, and specifications; to designate the particular technique to be used; and to verify the adequacy of procedures. The individual shall also have general familiarity with the NDT methods covered in Appendix A of

this standard. The NDT Level III shall be capable of conducting or directing the training and examining of NDT personnel in the methods for which the NDT Level III is qualified.

- 3.3 NDT Level II. An NDT Level II shall have the skills and knowledge to set up and standardize equipment, to conduct tests, and to interpret, evaluate, and document results in accordance with procedures approved by an NDT Level III. The NDT Level II shall be thoroughly familiar with the scope and limitations of the method to which certified and should be capable of directing the work of trainees and NDT Level I personnel. The NDT Level II shall be able to organize and report nondestructive test results.
- 3.4 NDT Level II Limited. An NDT Level II Limited shall have the skills and knowledge to set up and standardize equipment, to conduct tests, and to interpret, evaluate, and document results in accordance with procedures approved by an NDT Level III in the techniques listed in Appendix B. The NDT Level II Limited shall be thoroughly familiar with the scope and limitations of the technique to which certified and should be capable of directing the work of trainees and NDT Level I personnel. The NDT Level II Limited shall be able to organize and report nondestructive test results.
- 3.5 NDT Level I. An NDT Level I shall have the skills and knowledge to properly perform specific standardizations, specific tests, and, with prior written approval of the NDT Level III, perform specific interpretations and evaluations for acceptance or rejection and document the results, in accordance with specific approved procedures. The NDT Level I shall be able to follow approved nondestructive testing procedures and shall receive the necessary guidance or supervision from a certified NDT Level II or NDT Level III individual.
- 3.6 Trainee. A person who is not yet certified to any level shall be considered a trainee. Trainees shall work with a certified person, under the direction of an NDT Level II or NDT Level III and shall not independently conduct any tests or write a report of test results.
- 3.7 NDT Instructor. An NDT instructor shall have the skills and knowledge to plan, organize, and present classroom, laboratory, demonstration, and/or on-the-job NDT instruction, training, and/or education programs in accordance with course outlines approved by an NDT Level III.

4.0 Qualification Requirements

- 4.1 Training. Candidates for certification as NDT Level I or Level II shall complete sufficient organized training to become familiar with the principles of the method and the practices of the applicable test technique. This training shall be conducted in accordance with a course outline approved by an NDT Level III.
 - 4.1.1 NDT Level I and Level II. The minimum number of training hours required for NDT Level I and Level II candidates is described in Appendixes A and B. The course shall include the topics contained in ANSI/ASNT CP-105 for the appropriate NDT method, plus such additional topics as deemed necessary by the NDT Level III. The sequence, content, amount of time spent, and depth of coverage for each topic shall be approved by the NDT Level III. Training programs shall include sufficient examinations to demonstrate that the necessary information has been comprehended.
 - 4.1.1.1 The organized training may include instructor-led training, personalized instruction, virtual instructor-led training, computer-based training, or web-based training. Computer-based training and web-based training shall track hours and content of training with student examinations in accordance with paragraph 4.1.2.
 - 4.1.2 Credit. To receive credit for training hours, the individual shall pass a final examination covering the topics contained in that program.
 - 4.1.3 Presentation of Training. All training shall be presented by an NDT instructor. However, the NDT instructor may use personnel with specialized expertise (for example, metallurgists, welding engineers, etc.) who are not qualified in accordance with this standard to assist in presentation of specific information. The NDT Level III shall in all cases be responsible for the content of the completed course.
 - 4.1.4 NDT Level III. Training requirements for NDT Level III are satisfied as a result of holding a current ASNT Level III certificate in the specific NDT method.
- 4.2 Experience. Candidates for certification shall have acquired the practical experience to ensure they are capable of performing the duties of the level in which certification is being sought.
 - 4.2.1 NDT Level I and Level II. The minimum number of hours of experience required for NDT Level I and Level II candidates is described in Appendix A.
 - 4.2.2 NDT Level II Limited. The minimum number of hours of experience required for NDT Level II Limited candidates is described in Appendix B.
 - 4.2.3 NDT Level III. Experience requirements for NDT Level III are satisfied as a result of holding a current ASNT Level III certificate in the specific NDT method.

- 4.3 Previous Training and Experience
 - 4.3.1 NDT Level I and Level II. A candidate's previous training and experience may be accepted by the employer's NDT Level III if documented and verified. Any claimed training or experience that is not documented and cannot be verified shall be considered invalid.
 - 4.3.2 NDT Level III. The employer shall verify and document the current validity of a candidate's ASNT Level III certificate.
 - 4.4 NDT Instructor
 - 4.4.1 Criteria. An NDT instructor shall meet at least one of the following criteria:
 - 4.4.1.1 possess a current ASNT Level III certificate in the NDT method to be taught; or
 - 4.4.1.2 have academic credentials at least equivalent to a B.S. in engineering, physical science, or technology, and possess adequate knowledge in the NDT method to be taught; or
 - 4.4.1.3 be a graduate of a two-year school of science, engineering, or NDT and have five or more years of experience as an NDT Level II, or equivalent, in the NDT method to be taught; or
 - 4.4.1.4 have 10 or more years of NDT experience as an NDT Level II, or equivalent, in the NDT method to be taught.
 - 4.4.2 Designation. The NDT instructor shall be designated by an NDT Level III individual. The designation shall become part of the individual's qualification records.
 - 4.5 Outside Services. At the option of the employer, an outside organization may be engaged to perform the duties of an NDT Level III. For organizations other than ASNT, the employer shall be responsible for evaluating the organization to ensure the services are in accordance with the employer's certification procedure and this standard, and be so documented. An NDT Level III of the engaged outside organization shall be responsible for the services provided.
- 5.0 Qualification and Certification**
- 5.1 Certification Procedure. The employer shall develop and maintain a procedure detailing the program that will be used for qualification and certification of NDT personnel in accordance with this standard.
 - 5.2 Certification Procedure Requirements. The procedure shall describe the minimum requirements for certifying personnel in each NDT method and the levels of qualification desired. The procedure shall satisfy the requirements of this standard. The procedure shall include, as a minimum, the following:
 - 5.2.1 personnel duties and responsibilities;
 - 5.2.2 training requirements;
 - 5.2.3 experience requirements;
 - 5.2.4 examination requirements;
 - 5.2.5 records and documentation requirements, including control, responsibility, and retention period; and
 - 5.2.6 recertification requirements.
 - 5.3 Approval. The employer's certification procedure shall be approved by an NDT Level III designated by the employer.
- 6.0 Examinations**
- 6.1 Vision
 - 6.1.1 Near Distance. Prior to certification, NDT personnel shall be examined to ensure that they have natural or corrected (no pharmacological agents) near-distance acuity in at least one eye such that each individual is capable of reading Jaeger No. 1 test chart or equivalent* at a distance of not less than 12 in.
 *Equivalent eye examination results for Jaeger J-1 visual acuity are: Snellen 20/22; Times Roman 3.5 point text; OrthoRater #9; or Titmus (SAB-1, SAL-1 or SAR-1) #10.
 - 6.1.1.1 Pharmacological agents (eye drops) shall not be used which would improve or enhance visual acuity at any distance.
 - 6.1.2 Color Vision. NDT personnel for all methods shall demonstrate the ability to differentiate among the colors or shades of gray used in the method.
 - 6.1.3 Frequency. Vision examinations shall be administered annually, except that color differentiation examinations need be repeated only at each recertification. Vision examinations shall expire on the last day of the month of expiration.
 - 6.1.4 Administration. Vision examinations shall be administered in accordance with a procedure, and by personnel approved by an NDT Level III designated by the employer.

- 6.2 NDT Level III Examinations
- 6.2.1 Initial Requirement. Prior to the employer's certification examinations, the candidate shall hold a current ASNT Level III certificate with a currently valid endorsement for each method for which employer certification is sought.
- 6.2.2 Specific Examination (for each method). The employer shall administer a written examination consisting of at least 30 questions relating to the comprehension of the NDT-related requirements of specifications or standards used by the employer. Copies of the applicable specifications or standards shall be available as reference material during the examination.
- 6.2.3 Practical Examination. The candidate shall prepare an NDT procedure appropriate to the employer's needs; however, if documented experience demonstrates that the candidate has previously prepared acceptable NDT procedures in the method using the specifications, codes, and standards that are applicable to that employer, a written practical examination (for example, preparation of a procedure) is not required. If experience is substituted for the written practical examination, the employer shall document the pertinent practical experience of the NDT Level III candidate.
- 6.2.3.1 If the NDT Level III will be required to perform tests or evaluate test results, the practical examination shall include the same demonstrations of the candidate's ability to perform the required activity(ies) as required in paragraph 6.3.3.2.
- 6.3 NDT Level I and Level II Examinations
- 6.3.1 General. A General examination shall be approved by an NDT Level III. Administration of the examination shall be in accordance with paragraph 6.6 of this standard and shall be closed book. Reference material, such as charts, formulas, tables, and graphs, may be provided by the NDT Level III. Questions shall be developed which represent a cross section of the body of knowledge contained in ANSI/ASNT CP-105 applicable to each method and NDT level. The minimum number of questions required for each method and level is listed in Table 1. Questions used in General examinations for NDT Level I and Level II personnel shall be similar in type and difficulty to those published by ASNT.
- 6.3.1.1 A valid ACCP, ASNT NDT, or ASNT 9712 Level II certificate may be accepted as fulfilling the General examination criteria.
- 6.3.2 Specific. A Specific examination shall be approved by an NDT Level III. Administration shall be in accordance with paragraph 6.6 of this standard. The NDT Level III shall determine whether appropriate procedures, specifications, standards, or code sections will be provided. The examination shall address various examples of equipment, procedures, and test techniques that the candidate may use in the performance of assigned duties. The minimum number of questions required for each method and level is listed in Table 1.
- 6.3.2.1 A valid ACCP, ASNT NDT, or ASNT 9712 Level II certificate may be accepted as fulfilling the Specific examination criteria for each applicable method if the NDT Level III has determined that the ASNT examinations meet the requirements of the employer's certification procedure. This acceptance shall be documented. If this assessment cannot be accomplished, an employer-administered Specific examination shall be completed.
- 6.3.3 Practical. A practical examination shall be approved by an NDT Level III. Administration shall be in accordance with paragraph 6.6 of this standard. The practical examination shall consist of the following:
- 6.3.3.1 NDT Level I. The candidate shall demonstrate proficiency using the applicable nondestructive test method to examine at least one test sample for each technique to be used in the candidate's job and by documenting the results of the test. The test samples shall be representative of the products that the candidate will encounter in performing the job functions.
- 6.3.3.2 NDT Level II. The candidate shall demonstrate proficiency by performing the applicable nondestructive test method in examining at least one sample per technique and a minimum of two samples per method and by interpreting, evaluating, and documenting the results of the examination. The test samples shall be representative of the product that the candidate will encounter in performing the job functions.
- For purposes of the practical examinations described in paragraph 6.4.3, the techniques required to be demonstrated are in Appendix A. Where no techniques are listed in Appendix A, the employer must determine the techniques required for their particular needs in that method and specify those techniques required for certification in their qualification and certification procedure.

TABLE 1
Minimum Number of Examination Questions

Method/Technique	General		Specific	
	Level I	Level II	Level I	Level II
Acoustic Emission Testing	40	40	20	20
Electromagnetic Testing				
Alternating Current Field Measurement	40	40	20	20
Eddy Current Testing	40	40	20	20
Remote Field Testing	30	30	20	20
Leak Testing				
Bubble Leak Testing	20	20	15	15
Pressure Change Leak Testing	20	20	15	15
Halogen Diode Leak Testing	20	20	15	15
Mass Spectrometer Leak Testing	20	20	20	40
Liquid Penetrant Testing	40	40	20	20
Magnetic Flux Leakage Testing	20	20	20	15
Magnetic Particle Testing	40	40	20	20
Neutron Radiographic Testing	40	40	20	20
Radiographic Testing				
Radiography	40	40	20	20
Radiographic Interpretation – (Film, DR, CR, CT) Non-Radiographer		40		20
Radiographic Interpretation – (Film, DR, CR, CT) Radiographer (Certified NDT Level I)		20		15
Computed Radiography	40	40	20	20
Computed Tomography	40	40	20	20
Digital Radiography	40	40	20	20
Thermal/Infrared Testing	40		20	
Building Diagnostics		50		40
Electrical and Mechanical Testing		50		40
Materials Testing		50		40
Ultrasonic Testing	40	40	20	20
Time of Flight Diffraction		40		30
Phased Array Ultrasonic Testing		40		30
Full Matrix Capture		40		30
Ultrasonic Digital Thickness Measurement (numeric output only)		20		10
Ultrasonic Straight Beam (A-scan) Measurement		30		15
Vibration Analysis	40	40	20	60
Visual Testing	40	40	20	20

- 6.3.3.3 A valid ACCP, ASNT NDT, or ASNT 9712 Level II certificate may be accepted as fulfilling the practical examination criteria for each applicable method if the NDT Level III has determined that the ASNT examinations meet the requirements of the employer's certification procedure. This acceptance shall be documented. If this assessment cannot be accomplished, an employer-administered practical examination shall be completed.

6.4 Administration and Grading

6.4.1 Responsibilities

- 6.4.1.1 NDT Level I and II Examinations. An NDT Level III shall be responsible for the development, approval, administration, and grading of examinations for NDT Level I and Level II personnel for those methods in which the NDT Level III has a valid ASNT Level III certificate. For the practical examination, the individual administering the examination must be an NDT Level III in the respective test method.
- 6.4.1.2 NDT Level III Examinations. The employer shall be responsible for having an individual possessing a current ASNT Level III certificate, in the respective test method, develop, administer, and grade NDT Level III Specific and practical examinations.
- 6.4.1.3 The administration and grading of multiple-choice questions may be delegated by the NDT Level III, if so documented. Training shall be provided to the individual administering the examinations. This training and qualification shall be documented.

- 6.4.2 Employer Examinations. For each employer-administered certification written examination, each candidate shall achieve a grade of at least 70%. For practical examinations, each candidate shall achieve a grade of at least 80%. A composite average grade of 80% is required to be eligible for certification. All certification examinations shall have equal weight in determining the average grade.

6.4.3 Practical Examinations

- 6.4.3.1 For all practical examinations, all applicable discontinuities and/or conditions which the candidate is expected to find and/or evaluate must be specified and documented by the NDT Level III prior to administration of the examination.
- 6.4.3.2 A requirement for passing the NDT Level I, II, or III practical examination shall be the detection of the discontinuities and/or conditions previously specified by the NDT Level III responsible for preparing that examination.

- 6.4.4 NDT Level I Practical Examinations. The NDT Level III shall use a written checklist in administering and grading NDT Level I practical examinations. This checklist shall address at least the following items: proficiency in use of equipment and technique, proper adherence to procedure, test sequence, standardizations, materials, documentation, and extent of examinations. If the NDT Level I will interpret or evaluate results, the checklist shall include these item(s).

- 6.4.5 NDT Level II Practical Examinations. The NDT Level III shall use a written checklist in administering and grading NDT Level II practical examinations. The checklist shall address at least the following items: proficiency in use of techniques and equipment; proper adherence to procedure, test sequence, standardizations, and materials; satisfactory detection and location of discontinuities; proper extent of examination; and the accuracy and completeness of interpretations, evaluations, and documentation of the activities and test results.

- 6.4.5.1 Film Interpretation Limited Certification. The practical examination shall consist of a sufficient number of radiographs to demonstrate satisfactory performance to the satisfaction of the NDT Level III. The number of radiographs shall be addressed in the employer's certification procedure.
- 6.4.5.2 Phased Array Ultrasonic Testing (PAUT) and Time of Flight Diffraction (TOFD). Flawed samples used for practical examinations should be representative of the components and/or configurations that the candidates would be testing under this endorsement and approved by the NDT Level III.

- 6.4.6 NDT Level III Practical Examinations. Persons administering NDT Level III practical examinations shall use a written checklist. The checklist shall address items relating to the technical and practical adequacy of the NDT procedure(s) prepared by the candidate. When applicable to the candidate's job responsibilities, the checklist shall also address the items listed in paragraph 6.4.5.

- 6.5 Reexamination. Candidates who fail to attain the required passing grade must receive additional documented training or wait at least 30 days for reexamination. This training shall address the deficiencies which caused failure. A candidate shall not be reexamined using the examination or specimen previously failed or both.
- 6.6 Administration of Examinations. In no case shall an examination be prepared or administered by the individual being examined or by a subordinate.
- 6.7 Administration of NDT Level III Examinations. The employer's representative who administers the NDT Level III Specific and practical examinations shall possess a current ASNT Level III certificate in the method for which the examination is administered and shall be knowledgeable and familiar with the standards, specifications, and products used or made by the employer.
- 6.8 Certification. Certification of NDT personnel to all levels of qualification is the responsibility of the employer.
 - 6.8.1 Certification of NDT personnel is based on demonstrations of satisfactory qualification in accordance with paragraphs 4.1, 4.2, 6.1, 6.2, and 6.3 of this document, as described in the employer's certification procedure.
 - 6.8.2 Satisfactory completion of the qualification requirements shall be documented by the issuance of a certification record which has been signed by the NDT Level III who verified the qualifications have been met and the Certifying Authority issuing the certification record.
 - 6.8.3 Personnel certification records shall be maintained on file by the employer for the duration specified in the employer's written practice and shall contain those records identified in Section 9.0.
- 7.0 Expiration, Suspension, Revocation, and Reinstatement of Employer Certification
 - 7.1 Expiration. Individual certifications shall expire:
 - 7.1.1 when employment with the employer is terminated;
 - 7.1.2 on the last day of the month of expiration at the end of five years for NDT Level I and NDT Level II individuals;
 - 7.1.3 for NDT Level III individuals, when the ASNT Level III certificate has expired.
 - 7.2 Suspension. The employer shall suspend an individual's certification if:
 - 7.2.1 the near-distance vision examination interval exceeds one year. Certification is reinstated concurrently with passing the vision reexamination; or
 - 7.2.2 the individual has not performed the duties in the method(s) for which certified during any consecutive 12-month period; or
 - 7.2.3 the individual's performance is determined to be deficient in the required method or technique for specific documented reasons; or
 - 7.2.4 for NDT Level III personnel, when the ASNT Level III certificate has not been renewed.
 - 7.3 Revocation. The employer shall revoke an individual's certification when:
 - 7.3.1 the individual has not performed the duties in the method(s) for which certified during any consecutive 24-month period; or
 - 7.3.2 for NDT Level III personnel, the ASNT Level III certificate has been revoked; or
 - 7.3.3 an individual's conduct is deemed by the employer to be or have been unethical or incompetent.
 - 7.4 Reinstatement
 - 7.4.1 Suspended Certifications. Reinstatement of suspended certifications for NDT Level I or NDT Level II shall be determined by the NDT Level III. Reinstatement of suspended NDT Level III certifications shall be determined by the employer, except that the requirement for ASNT Level III certification may not be waived.
 - 7.4.2 Expired or Revoked Certifications. Certifications which have expired or have been revoked may only be reinstated by complying with Section 6.0, paragraph 7.4.3, or paragraph 8.1. Reinstatement per paragraph 7.4.3 is only allowed when the employee's certification expired due to termination of employment. Such reinstatement of certification shall only be for the period of time remaining on the original certification.
 - 7.4.3 Expired for Termination of Employment. An NDT Level I, Level II, or Level III whose certification has expired due to termination of employment may be reinstated to the former NDT level, without a new examination, providing all of the following conditions are met:
 - 7.4.3.1 the employer has maintained the personnel certification records required in Section 9.0;
 - 7.4.3.2 the employee certification did not expire during termination; and
 - 7.4.3.3 the employee is being reinstated within 12 months of termination.

8.0 Employer Recertification

- 8.1 NDT Level I and NDT Level II. NDT Level I and NDT Level II personnel shall be recertified by reexamination in accordance with Section 6.0 or by meeting the requirements of paragraph 8.1.1, when applicable.
 - 8.1.1 Every five years, the individual may be recertified if the individual has at least 350 hours of documented experience using the applicable method over the five-year certification interval and has successfully passed a Specific examination which complies with paragraph 6.3.2.
 - 8.1.2 At least every 10 years, the individual must repeat the examinations in Section 6.0.
- 8.2 NDT Level III. NDT Level III personnel shall be recertified by the employer every five years by verifying that the individual's ASNT Level III certificate is current in each method for which recertification is sought.

9.0 Records

- 9.1 Responsibility for Documentation. The employer shall document certifications in accordance with this standard.
- 9.2 Contents of Documentation. The employer's certification documentation shall include at least a training record, a certification record, an experience record, a record of previous experience (if applicable), employee's current examinations, and a vision examination record.
 - 9.2.1 Certification Record. The certification record shall include at least the following information:
 - 9.2.1.1 name of certified individual;
 - 9.2.1.2 level of certification and NDT method, including the test technique covered; limitations (if any), as applicable;
 - 9.2.1.3 results of current employer examinations that the individual has taken;
 - 9.2.1.4 for Level III personnel, a copy of the candidate's ASNT Level III certificate;
 - 9.2.1.5 dates of certification, expiration, suspension, revocation, and reinstatement; and
 - 9.2.1.6 the signature, printed name, and title of both the designated NDT Level III that verified the qualifications of candidate for certification and the signature of employer's Designated Certifying Authority, when the Level III and Certifying Authority are identified as separate positions in the certification procedure.
 - 9.2.2 Dates of Certification and Expiration.
 - 9.2.2.1 The date of certification shall be a date after which the employer's requirements for education, training, experience, and examinations have been met for each method and/or technique, as specified in the employer's certification procedure.
 - 9.2.2.2 The date of certification is the date when the Certifying Authority signs the certification record.
 - 9.2.2.3 The date of expiration should be five years from the date established in paragraph 9.2.2.2.
 - 9.2.3 NDT Training Record. A documented history of the employee's training shall be maintained that identifies NDT training received by the individual, the organization providing the training, dates of the training, hours of training, evidence of satisfactory completion, and the instructor's name.
 - 9.2.4 NDT Experience Record. A record that identifies the individual's experience performing various nondestructive tests shall be maintained for purposes of verifying initial certification experience and continuing experience.
 - 9.2.5 Record of Previous Experience. Documented evidence of the individual's previous NDT training and experience shall be maintained if previous training and experience are used to satisfy any part of the requirements of this standard.
 - 9.2.6 Visual Examination Records. Current records of vision examinations required by paragraph 6.1 shall be maintained.
 - 9.2.7 Certification Certificates and Wallet Cards. The following requirements shall be followed if certification certificates and/or wallet cards are issued by the employer:
 - 9.2.7.1 When certificates or wallet cards are issued and provided by the employer, the requirements of ASNT Policy G-14 (asnt.org) shall be followed to ensure appropriate usage of the ASNT name, acronym, and logo.
 - 9.2.7.2 The certificate shall indicate that the certification is in accordance with the requirements of the employer's certification procedure.
 - 9.2.7.3 The legal name of the employer shall be clearly identified followed by the document number and revision/edition number of the certification procedure.
 - 9.2.7.4 The certificate shall indicate "CP-189" followed by the four-digit year of the applicable edition of CP-189 that the certification procedure was developed to follow.

- 9.2.7.5 Full legal name of the certified individual in the ISO basic Latin alphabet. Additional characters may also be used where customary to accommodate local languages.
- 9.2.7.6 Level of certification, NDT method, and/or technique, as applicable and limitations (if any).
- 9.2.7.7 Limitations of certification authorization should be clearly defined, when appropriate for the employer, which may include restrictions on product forms, industry sectors, codes, standards, and examination procedures.
- 9.2.7.8 The full printed legal name and signature of both the designated NDT Level III that verified the qualifications of the candidate for certification and the designated Certifying Authority when the Level III and Certifying Authority are identified as separate positions in the employer's certification procedure.
 - 9.2.7.8.1 The ASNT-issued Level III identification number, when either or both of the signers hold ASNT Level III certifications.
- 9.2.7.9 The certificate or wallet card should be fully completed at the time of signing by the designated NDT Level III and/or Certifying Authority so that no fill-in-the-blank data remains to be completed by others.
- 9.2.7.10 When changes or alterations to certificates or wallet cards are necessitated, this requires the issuing of new certificates or wallet cards.
- 9.2.7.11 Dates of certification and expiration.

10.0 Referenced Publications

- 10.1 The following documents contain provisions which, through reference in this text, constitute provisions of this standard. Copies may be obtained from The American Society for Nondestructive Testing Inc., 1201 Dublin Road, Suite #G04, Columbus, OH 43215, USA, asnt.org.
 - 10.1.1 ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel*, latest edition
 - 10.1.2 ASNT NDT Level III program, *ASNT Document CP-ASNT*, latest edition
 - 10.1.3 ASNT Central Certification Program, *ASNT Document CP-1*, latest edition
 - 10.1.4 *ASNT 9712 Certification Program, Qualification and Certification of NDT Personnel to ISO 9712*, latest edition
 - 10.1.5 ASNT Policy G-14 – *Use of the ASNT Name and ASNT Marks*, latest edition
- 10.2 The following document contains specific NDT terms which, though referenced in this text, constitute provisions of this standard. Copies may be obtained from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, USA, astm.org.
 - 10.2.1 ASTM E1316 (latest edition) – *Standard Terminology for Nondestructive Examinations, Section A - Common NDT Terms*

APPENDIXES

Appendix A

Initial Training and Experience Requirements for Level I and Level II

Evaluation Method	Level	Technique	Required Training (Hours)	Required Experience	
				Minimum Hours in Method or Technique	Total Hours in NDT
Acoustic Emission Testing (AE)	I		40	210	400
	II		40	630	1200
Electromagnetic Testing (ET)	I	Alternating Current Field Measurement (ACFM)	40	210	400
	II		40	630	1200
	I	Eddy Current Testing (ECT)	40	210	400
	II		40	630	1200
	I	Remote Field Testing (RFT)	40	210	400
	II		40	630	1200
Leak Testing (LT)	I	Bubble Leak Testing (BT)	2	3	15
	II		4	35	80
	I	Pressure Change Leak Testing (PCT)	24	105	200
	II		16	280	530
	I	Halogen Diode Leak Testing (HDLT)	12	105	200
	II		8	280	530
	I	Mass Spectrometer Leak Testing (MST)	40	280	530
	II		24	420	800
Liquid Penetrant Testing (PT)	I		4	70	130
	II		8	140	270
Magnetic Flux Leakage Testing (MFL)	I		16	70	130
	II		12	210	400
Magnetic Particle Testing (MT)	I		12	70	130
	II		8	210	400
Neutron Radiographic Testing (NR)	I		28	420	800
	II		40	1680	2400
Radiographic Testing	I	Radiography (RT)	40	210	400
	II		40	630	1200
	I	Computed Radiography (CR)	40	210	400
	II		40	630	1200
	I	Computed Tomography (CT)	40	210	400
	II		40	630	1200
	I	Digital Radiography (DR)	40	210	400
	II		40	630	1200

(CONTINUED NEXT PAGE)

Appendix A (continued from page 11)**Initial Training and Experience Requirements for Level I and Level II**

Evaluation Method	Level	Technique	Required Training (Hours)	Required Experience	
				Minimum Hours in Method or Technique	Total Hours in NDT
Thermal/Infrared Testing (TIR)	I		32	210	400
	II	Building Diagnostics	34	1260	1800
	II	Electrical & Mechanical Testing	34	1260	1800
	II	Materials Testing	34	1260	1800
Ultrasonic Testing (UT)	I		40	210	400
	II		40	630	1200
	II	Time of Flight Diffraction (TOFD)	40	320	n/a
	II	Phased Array Ultrasonic Testing (PAUT)	80	320	n/a
	II	Full Matrix Capture (FMC)	80	320	n/a
Vibration Analysis (VA)	I		24	420	800
	II		72	1680	2400
Visual Testing (VT)	I		8	70	130
	II		16	140	270

NOTES:

- A: Experience shall be based on the actual hours worked in the specific method or technique.
- B: A person may be qualified directly to NDT Level II with no time as a certified NDT Level I, providing the required training and experience consists of the sum of the hours required for NDT Level I and NDT Level II.
- C: The required minimum experience shall be documented by method and by hour with supervisor or NDT Level III approval.
- D: While fulfilling total NDT experience requirements, experience may be gained in more than one method; however, the minimum hours must be met for each method.
- E: To qualify in a single method, the hours in the Total Hours column must be satisfied.
- F: If an individual is certified in one ET technique, the minimum experience to qualify in another ET technique at the same level or to the next level may be reduced by up to 50%.
- G: If an individual is currently certified in a radiographic technique and a full-course format was used to meet the initial qualifications in that technique, the minimum training hours to qualify in another technique at the same level shall be 24 hours (of which at least 16 hours shall be equipment familiarization) as defined in the employer's certification procedure. If an individual is certified in a technique, the minimum additional experience to qualify for another technique at the same level may be reduced by up to 50%, as defined in the employer's certification procedure.
- H: FMC, TOFD, and PAUT require UT Level II certification as a prerequisite.
- I: If an individual is currently certified in one thermal/infrared technique and a full-course format was used to meet the initial qualifications in that technique, the minimum additional training hours to qualify in another technique at the same level shall be 20 hours (of which at least 16 hours shall be specific technique familiarization). The training outline shall be as defined in the employer's certification procedure. If an individual is certified in a technique, the minimum additional experience required to qualify for another technique at the same level may be reduced by up to 50%, as defined in the employer's certification procedure.
- J: Independent of the RT, CR, CT, DR, or NR training required for Level I and Level II certification, a trainee is required to receive radiation safety training as required by the regulatory jurisdiction.
- K: In addition to the training required in this table for FMC, TOFD, and PAUT, supplemental specific hardware and software training should be required for automated or semiautomated technique applications. The employer's certification procedure shall fully describe the nature and extent of the additional training required for each specific acquisition or analysis software and instrument/system used. The employer's certification procedure shall also describe the means by which the examiner's qualification will be determined for automated and semiautomated techniques.

Appendix B

Initial Training and Experience Requirements for NDT Level II Limited Certifications

Evaluation Technique	Required Training (Hours)	Required Experience Minimum Hours in the Technique
Radiographic Image Interpretation - Non-Radiographer Film, DR, CR, CT	40	220
Radiographic Image Interpretation - Certified NDT Level I Film, DR, CR, CT	24	220
Ultrasonic Digital Thickness Measurement (Numeric output only)	8	40
Ultrasonic Straight Beam (A-scan) Thickness Measurement	24	175

(Not a part of American National Standard ANSI/ASNT CP-189)

Appendix C – Sample Certification Form

Note: The form below is an example and may be revised to meet the employer's requirements.

[COMPANY NAME]**NONDESTRUCTIVE TESTING PERSONNEL CERTIFICATION RECORD**

Name:	I.D. Number:
Current Certification Status: ☐ Level I ☐ Limited ☐ Level II ☐ Level III	
Method:	Recertification Due Date:
Recertified by Continuing Satisfactory Experience: Date _____ By _____	

Education

High School or Equivalent: _____ Advanced: _____

Certificate(s) and/or Degree(s) Obtained/Date(s): _____

NDT Training

Course	Date	Hours	Instructor

Experience

Employer	Method(s)	Dates (from/to)	Months or Hours

Qualification Examination Results

General	Specific	Practical	Composite Score	Date	Administered By

The above-named employee has satisfactorily met the qualification requirements of the [INSERT COMPANY NAME] certification procedure in accordance with the 2024 edition of CP-189. Other details of qualification, if any, are maintained on file and are available for audit upon request.

Printed Name: _____ Title: _____

Signed: _____ Date: _____

Appendix D – Sample Certificate

Date of Certification:	{Insert}
Date of Certification Expiration:	{Insert}

9.2.7.1

{INSERT LEGAL COMPANY NAME}

9.2.7.3

NDT CERTIFICATE

This is to certify that the individual named below has successfully completed experience, training, and examination requirements in accordance with the provisions of {INSERT LEGAL COMPANY NAME} NDT Certification Procedure for the Qualification and Certification of NDT personnel {INSERT CERTIFICATION PROCEDURE IDENTIFICATION NUMBER & REVISION}.

9.2.7.3

9.2.7.5

{INSERT FULL LEGAL NAME OF INDIVIDUAL}

Is hereby certified to perform the following Nondestructive Testing Method(s)

9.2.7.6 & 9.2.7.7

NDT Method	NDT Level	NDT Technique (as applicable per certification procedure)	Limitations (if any)
{Insert}	{Insert}	{Insert}	{Insert}

9.2.7.6

9.2.7.6

9.2.7.6

Certified on behalf of the Certifying Authority - {INSERT LEGAL COMPANY NAME}

Sign:

9.2.7.8 & 9.2.7.8.1

Print Name:

{INSERT LEGAL COMPANY NAME} Designated Level III

Sign:

9.2.7.8

Print Name:

9.2.7.4

This company NDT Certification Procedure intends to meet or exceed the requirements of ANSI/ASNT CP-189 {INSERT EDITION YEAR OF ISSUANCE THAT THE COMPANY IS REQUIRED TO COMPLY WITH} as it applies to NDT performed by this company.

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